

Formoterol on airway smooth muscle and human lung mast cells: a comparison with salbutamol and salmeterol.

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Formoterol, like salbutamol and salmeterol, relaxed isolated preparations of guinea-pig trachea and human bronchus, and inhibited antigen-induced mediator release from human lung fragments in a concentration-related fashion. In each case, these actions were mediated through beta 2-adrenoceptors, with formoterol being 50-120-fold more potent than salbutamol, and 2-27-fold more potent than salmeterol. The duration of action of formoterol was longer than that of salbutamol in all preparations, but was markedly shorter than that of salmeterol, whose actions persisted for many hours despite continuous or extensive washing of the tissues. In conscious guinea-pigs, inhaled formoterol, salbutamol and salmeterol all caused dose-related inhibition of histamine-induced bronchoconstriction. Formoterol was again more potent (10-20-fold) than either salbutamol or salmeterol. However, while the actions of a threshold-effective dose of formoterol persisted for less than 3 h, somewhat longer than those of salbutamol (< 1.5 h), an equivalent dose of salmeterol was active for at least 6 h. Therefore, while formoterol is a potent beta 2-adrenoceptor agonist in vitro and in vivo, and is consistently longer-acting than salbutamol, its duration of action is markedly shorter than that of salmeterol.

[Onset of action of formoterol in patients with moderate to severe, partially reversible airflow obstruction assessed by bodyplethysmography].

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Formoterol is a long acting beta2-agonist used for the treatment of obstructive airway diseases. Compared with Salmeterol, Formoterol has a rapid onset of bronchodilation. There are only scant data regarding the comparative onset of action using bodyplethysmography in moderate to severely obstructive patients. In a mono-center, single-blinded parallel group study 60 patients (age: 61.9 +/- 12.8 years, 65 % male) with moderate to severe (mean FEV(1) 40.6 +/- 15.3 % of predicted), partially reversible (FEV(1) post bronchodilator > 15 % from baseline) airway obstruction were treated with either formoterol 12 microg bid or salmeterol 50 microg bid over a period of two weeks. Onset of action was measured by airway resistance (sRaw) before and after two weeks of treatment. Compared with Salmeterol, Formoterol had a significantly faster onset of action (10 % decrease of Raw) at baseline (F: 1.4 +/- 0.9 vs. S: 15.1 +/- 34.5 min., $p < 0.0001$) and after two weeks of treatment (F: 6.2 +/- 21.6 vs. S: 51 +/- 135 min., $p < 0.0001$). Morning FEV(1) improved similarly during treatment in both study groups, when compared with baseline lung function (F: 1.38 +/- 0.64 vs. 1 +/- 0.41 l; S: 1.43 +/- 0.67 vs. 1 +/- 0.4 l, $p < 0.05$, both comparisons). Both treatments were well tolerated. Formoterol produces a rapid improvement of airway resistance in patients with moderate to severe, partially reversible airway obstruction. The onset of bronchodilation was significantly faster for Formoterol compared with Salmeterol. Both drugs improved lung function similarly after two weeks of treatment. It is important to distinguish beta2-agonists not only into short- and long-acting but also into fast- and slow-acting.

Treatment of childhood asthma: how do the available options compare?

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The National Asthma Council of Australia suggests that "the aim of preventive therapy should be to enable patients to enjoy a normal life (comparable with that of non-asthmatic children), with the least amount of medication and at minimal risk of adverse events. The level of maintenance therapy should be determined by symptom control and lung function in the interval periods." The British Thoracic Society/Scottish Intercollegiate Guidelines Network states that the aims of the pharmacological treatment of asthma should be to control symptoms, prevent exacerbations and achieve the best possible lung function with minimal adverse effects. We have used the current published international guidelines to highlight the international differences in management recommendations, and compared the possible pharmacological options with a focus on the above ideals. Cromones have been used for many years in childhood asthma. Most evidence suggests they now have little role. Regarding inhaled corticosteroids (ICS), beclomethasone and budesonide are essentially similar in their efficacy. Fluticasone propionate is equally as effective at one-half the equivalent dose of budesonide or beclomethasone. Adverse effects are rare in dosages <400 microg/day of budesonide and beclomethasone or <200 microg/day of fluticasone propionate, but may occur in individual patients. Relevant clinical adverse effects are rare and pharmacological systemic effects are less noticeable with budesonide and fluticasone propionate than with beclomethasone, but data are conflicting. Long-acting beta2-adrenoceptor agonists (beta2-agonists) are recommended once low-dose ICS have failed to control symptoms. The main pharmacological difference between the agents is that formoterol is a full beta2-adrenergic agonist, whereas salmeterol is a partial agonist at the beta2-adrenoceptor and has a unique pharmacological action. The main clinical distinction between these two agents is that their onset of bronchodilation differs. Bronchodilation begins at about 3 minutes after inhalation of formoterol, which is similar to the short-acting agents, whereas salmeterol has a much slower onset of action at about 15-30 minutes. The many in vitro differences between the two drugs are probably not clinically relevant. There are no comparative pediatric data on the leukotriene modifiers to make clear recommendations.

Inhaled dry-powder formoterol and salmeterol in asthmatic patients: onset of action, duration of effect and potency.

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Salmeterol and formoterol are two long-acting beta2-agonists for inhalation, currently being used in clinical practice. The aim of the present study was to investigate the onset of action, duration of effect and potency of these two beta2-agonists in asthmatic patients. Patients (n=28) were included on the basis of salbutamol stepwise reversibility (100, 100 and 200 microg, given cumulatively; total reversibility > or =15%). In a double-blind placebo-controlled crossover study, the bronchodilating properties of formoterol 6, 12 and 24 microg were compared with the effects of salmeterol 50 microg. Formoterol was given via Turbuhaler and salmeterol via Diskhaler, and forced expiratory volume in one second (FEV1) was monitored during 12 h. Formoterol at all doses had a more rapid onset than salmeterol as judged from bronchodilation at 3 min after the dose. Formoterol at all doses had a similar duration of effect to salmeterol 50 microg, as judged from bronchodilation at 12 h after dose administration. When the relative potency of the two drugs was compared, salmeterol 50 microg was estimated to correspond to formoterol 9 microg (95% confidence interval: 3-19 microg). We confirm that formoterol and salmeterol are both long-acting beta2-agonists, but with some differences in effect profile. We confirm the more rapid onset of action of formoterol compared with salmeterol, and furthermore, no difference in duration of effect is evident.

Onset of action and duration of effect of formoterol and salmeterol compared to salbutamol in isolated guinea pig trachea with or without epithelium.

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Formoterol and salmeterol are 2 newly developed beta 2-adrenoceptor agonists for inhalation with prolonged duration of effect compared with currently available beta 2-agonists. The clinical duration of effect has been suggested to be correlated to a decreased washability in vitro, i.e. decreased effect due to continuous washing of the organ bath. In this study we compared the washability and the onset of the relaxatory effect of formoterol (0.01 microM), salmeterol (0.05 microM) and salbutamol (0.1 microM) in isolated guinea pig trachea contracted with carbachol (0.1 microM). We also evaluated a possible influence of the epithelium on onset of action and/or duration of effect. A significant relaxatory effect of formoterol and salmeterol remained after washing, whereas no effect of salbutamol remained. We also found a rapid onset of action for both salbutamol and formoterol, but a significantly slower onset for salmeterol. Both the washability and the onset of action were found to be independent of the presence of the epithelium.

Studies on the interaction between formoterol and salmeterol in guinea-pig trachea in vitro.

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The actions of and interaction between formoterol and salmeterol were studied on guinea-pig trachea in vitro. Tracheal strip preparations were contracted by 1 mumol/l carbachol giving a near maximal contraction. Salmeterol in concentrations from 0.1 to 3 mumol/l relaxed the tracheal smooth muscle by about 30 per cent of the maximum relaxation produced by theophylline. Formoterol caused a concentration-dependent and almost complete relaxation with a pD₂ of 8.56. In the presence of salmeterol there was a rightward shift of the concentration-response curve for formoterol. The pA₂ for salmeterol was estimated to 7.42. Similar experiments with isoprenaline indicated that salmeterol has a low affinity for beta 1-adrenoceptors. Formoterol and salmeterol both inhibited in a concentration-dependent manner the contractions evoked by stimulation of the vagus nerve in a tracheal tube preparation. The degree of inhibition decreased with increasing stimulation frequency. Complete inhibition was attained with salmeterol, but not with formoterol, at the highest frequency employed (45 Hz). The inhibiting effect of 10 mumol/l salmeterol was not blocked by 10 mumol/l sotalol, a beta-adrenoceptor antagonist. It is concluded that salmeterol, in comparison to formoterol, is a partial beta 2-adrenoceptor agonist and has, at high concentrations, an additional unspecific inhibitory action.

Acute protection against exercise-induced bronchoconstriction by formoterol, salmeterol and terbutaline.

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The onset of bronchoprotection as obtained by various beta2-agonists has not been examined in a comparative study. In this study, the onset of bronchodilation and protection against exercise-induced bronchoconstriction in asthmatics after inhalation of the long-acting beta2-agonists formoterol and salmeterol and the short-acting beta2-agonist terbutaline were measured. Twenty-five subjects with

asthma and a history of exercise-induced bronchoconstriction (mean baseline forced expiratory volume in one second (FEV1): 90% predicted; mean fall in FEV1 after exercise: 31% from baseline) were enrolled in this double-blind, double-dummy, placebo-controlled, randomized, four-period crossover study. Exercise challenges were performed on 12 days at either 5, 30, or 60 min after inhalation of a single dose of formoterol (12 microg Turbuhaler), salmeterol (50 microg Diskus), terbutaline (500 microg Turbuhaler) or placebo. Exercise-induced bronchoconstriction (maximum fall in FEV1 or area under the curve) did not differ significantly between terbutaline, formoterol and salmeterol either 5, 30, or 60 min after inhalation of the study medication. In contrast, the onset of bronchodilation was slower after salmeterol compared to terbutaline and formoterol ($p < 0.05$, each), which both showed a similar time course. At all time points between 5 and 60 min, formoterol provided significantly greater bronchodilation than salmeterol ($p < 0.05$). These data indicate that equipotent doses of the bronchodilators salmeterol, formoterol and terbutaline were similarly effective with respect to their short-term protective potency against exercise-induced bronchoconstriction, despite the fact that the time course of bronchodilation was significantly different between the three beta2-agonists.

Decreased bronchodilating effect of salbutamol in relieving methacholine induced moderate to severe bronchoconstriction during high dose treatment with long acting beta2 agonists.

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In vitro the long acting beta2 agonist salmeterol can, in contrast to formoterol, behave as a partial agonist and become a partial antagonist to other beta2 agonists. To study this in vivo, the bronchodilating effect of salbutamol was measured during methacholine induced moderate to severe bronchoconstriction in patients receiving maintenance treatment with high dose long acting beta2 agonists. A randomised double blind crossover study was performed in 19 asthmatic patients with mean forced expiratory volume in one second (FEV1) of 88.4% predicted and median concentration of methacholine provoking a fall in FEV1 of 20% or more (PC(20)) of 0.62 mg/ml at entry. One hour after the last dose of 2 weeks of treatment with formoterol (24 microg twice daily by Turbuhaler), salmeterol (100 microg twice daily by Diskhaler), or placebo a methacholine provocation test was performed and continued until there was at least a 30% decrease in FEV1. Salbutamol (50 microg) was administered immediately thereafter, followed by ipratropium bromide (40 microg) after a further 30 minutes. Lung function was monitored for 1 hour after provocation. There was a significant bronchodilating and bronchoprotective effect after 2 weeks of active treatment. The dose of methacholine needed to provoke a fall in FEV1 of $\geq 30\%$ was higher after pretreatment with formoterol (2.48 mg) than with salmeterol (1.58 mg) or placebo (0.74 mg). The difference between formoterol and salmeterol was statistically significant: 0.7 doubling dose steps (95% CI 0.1 to 1.2, $p = 0.016$). The immediate bronchodilating effect of subsequently administered salbutamol was significantly impaired after pretreatment with both drugs ($p < 0.0003$ for both). Three minutes after inhaling salbutamol the increase in FEV1 relative to the pre-methacholine baseline was 15.8%, 7.3%, and 5.5% for placebo, formoterol and salmeterol, respectively (equivalent to increases of 26%, 14%, and 12%, respectively, from the lowest FEV1 after methacholine). At 30 minutes significant differences remained, but 1 hour after completing the methacholine challenge FEV1 had returned to baseline values in all three treatment groups. Formoterol has a greater intrinsic activity than salmeterol as a bronchoprotective agent, indicating that salmeterol is a partial agonist compared with formoterol in contracted human airways in vivo. Irrespective of this, prior long term treatment with both long acting beta2 agonists reduced the bronchodilating effect of an additional single dose of salbutamol equally, indicating that the development

of tolerance or high receptor occupancy overshadowed any possible partial antagonistic activity of salmeterol. Patients on regular treatment with long acting beta2 agonists should be made aware that an additional single dose of a short acting beta2 agonist may become less effective.

Long-acting inhaled beta2-agonists in asthma therapy.

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To review the pharmacology of the long-acting inhaled beta2-agonists, salmeterol and formoterol, summarize results of their clinical trials, evaluate their safety records, and discuss their roles in the treatment of asthma. Preclinical and clinical studies involving salmeterol or formoterol were identified by a MEDLINE search, weekly computerized literature updates, and manual searches. Studies of satisfactory quality were chosen for review. Salmeterol and formoterol are potent and selective beta2-adrenoceptor agonists with durations of action >12 h. Their major differences are that formoterol has a rapid onset of action and is a partial agonist of high intrinsic efficacy, whereas salmeterol has a delayed onset and is a partial agonist of low intrinsic efficacy. Twice daily use of either drug results in improved lung function, reduced symptoms, and a better quality of life. These agents protect against exercise-induced asthma for 12 h and eliminate nighttime awakening in most patients. Limited tolerance develops, especially to their bronchoprotective effects, but their improvement of lung function is sustained. Regular use of salmeterol or formoterol provides subjective and objective amelioration of asthma in patients experiencing excessive symptoms or physiologic impairment despite the regular administration of low doses of inhaled corticosteroids (equivalent to approximately 500 microg/d of beclomethasone). Intermittent use of either long-acting beta2-agonist can provide prolonged protection against exercise-induced asthma or nighttime symptoms. Patients should be instructed to continue taking inhaled steroids when long-acting beta2-agonists are administered on a regular schedule and to not take long-acting beta2-agonists between regularly scheduled doses. Used properly, they are effective and safe adjunctive agents in the treatment of asthma.

Elevated developmental temperatures below the lethal limit reduce *Aedes aegypti* fertility.

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Aedes aegypti mosquitoes are the principal vectors of dengue and continue to pose a threat to human health, with ongoing urbanization, climate change and trade all impacting the distribution and abundance of this species. Hot periods are becoming increasingly common and their impacts on insect mortality have been well established, but they may have even greater impacts on insect fertility. In this study, we investigated the impacts of high temperatures on *Ae. aegypti* fertility both within and across generations. Mosquitoes developing under elevated temperatures exhibited higher critical thermal maxima (CT_{max}), reflecting developmental acclimation, but their fertility declined with increasing developmental temperature. In females, elevated developmental temperatures decreased fecundity while in males it tended to decrease the proportion of eggs that hatched and the proportion of individuals producing viable offspring. Rearing both sexes at 35°C increased fecundity in the subsequent generation but effects of elevated temperatures persisted across gonotrophic cycles within the same generation. Moreover, exposure of adults to 35°C further decreased fertility beyond the effects of developmental temperature

alone. These findings highlight sub-lethal impacts of elevated temperatures on *Ae. aegypti* fertility and plastic responses to thermal stress within and across generations. This has significant implications for predicting the distribution and abundance of mosquito populations thriving in increasingly warmer environments.
